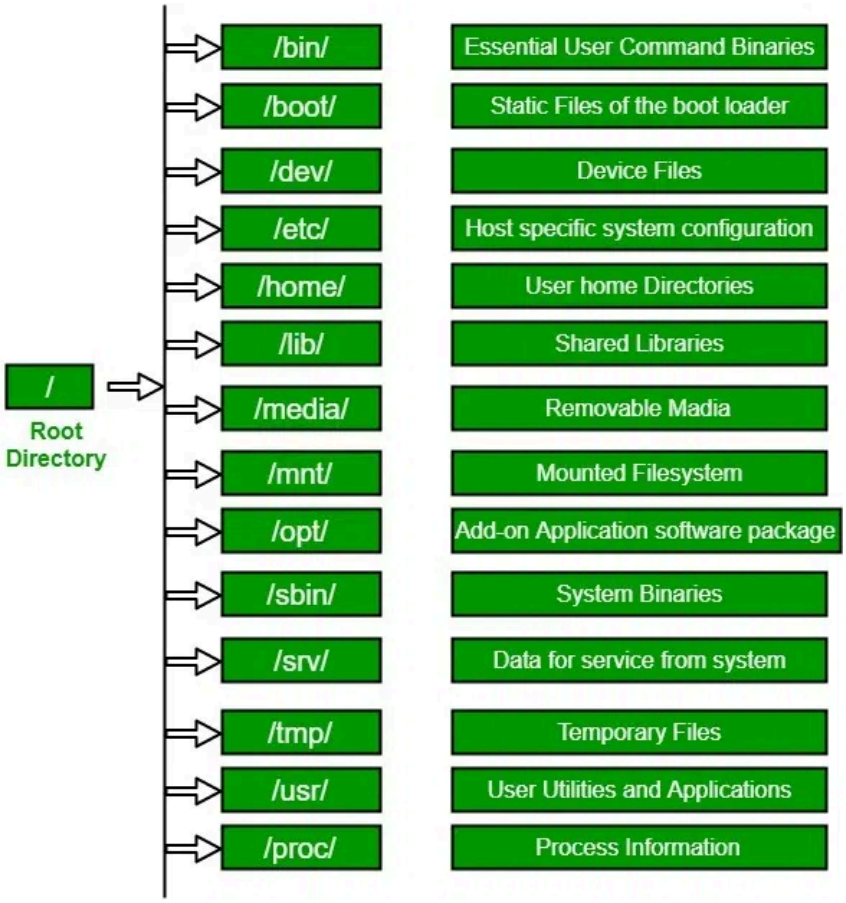
3Explore Linux Filesystem hierarchy.



/ – The root directory

Everything, all the files and directories, in Linux are located under ‘root’ represented by ‘/’. If you look at the directory structure, you’ll realize that it is similar to a plant’s root.

```
   65  history
root@apparent-mantis:/home# cd /
root@apparent-mantis:/# ls
bin                 etc                 lib64       opt   sbin                sys
bin.usr-is-merged   home                lost+found  proc  sbin.usr-is-merged  tmp
boot                lib                 media       root  snap                usr
dev                 lib.usr-is-merged   mnt         run   srv                 var
root@apparent-mantis:/#
```

Bin:
/bin : This directory contains executable programs which are needed in single user mode and to bring the system up or repair it.

```
root@ip-172-31-36-123:/# ls
bin  boot  dev  etc  home  lib  lib32  lib64  libx32  lost+found  media  mnt  opt  proc  root  run  s
root@ip-172-31-36-123:/# cd /bin
root@ip-172-31-36-123:/bin# ls -lsah
total 114M
  36K drwxr-xr-x  2 root root     36K May 17 04:56  .
 4.0K drwxr-xr-x 14 root root    4.0K Apr 11 02:07  ..
    0 lrwxrwxrwx  1 root root       4 Feb 17  2020  NF -> col1
 128K -rwxr-xr-x  1 root root    127K Dec  5 21:18  VGAuthService
    0 lrwxrwxrwx  1 root root       1 Mar 25  2022  X11 -> .
  52K -rwxr-xr-x  1 root root     51K Feb  8 03:46 '['
  36K -rwxr-xr-x  1 root root     35K Jun  5  2023  aa-enabled
  36K -rwxr-xr-x  1 root root     35K Jun  5  2023  aa-exec
  32K -rwxr-xr-x  1 root root     31K Jun  5  2023  aa-features-abi
  60K -rwxr-xr-x  1 root root     59K Apr 10 17:45  ab
  20K -rwxr-xr-x  1 root root     19K Jan 25  2022  acpi_listen
  16K -rwxr-xr-x  1 root root     15K Oct 24  2023  add-apt-repository
  16K -rwxr-xr-x  1 root root     15K Apr  9 15:32  addpart
    0 lrwxrwxrwx  1 root root      26 Jan 23 15:08  addr2line -> x86_64-linux-gnu-addr2line
 4.0K -rwxr-xr-x  1 root root    2.6K Apr 13  2023  apport-bug
  16K -rwxr-xr-x  1 root root     14K Apr 13  2023  apport-cli
    0 lrwxrwxrwx  1 root root      10 Apr 13  2023  apport-collect -> apport-bug
 4.0K -rwxr-xr-x  1 root root    2.1K Apr 13  2023  apport-unpack
  16K -rwxr-xr-x  1 root root     15K Mar 25  2022  appres
```

/boot : Contains static files for the boot loader. This directory only holds the files which are needed during the boot process.

```
System.map-6.5.0-1020-aws  config-6.5.0-1020-aws  grub  initrd.img-6
root@ip-172-31-36-123:/boot# ls -lsah
total 127M
4.0K drwxr-xr-x  4 root root 4.0K May 15 06:14 .
4.0K drwxr-xr-x 19 root root 4.0K May 30 04:47 ..
8.0M -rw-------  1 root root 8.0M Mar 25 17:52 System.map-6.5.0-1017
8.0M -rw-------  1 root root 8.0M May  1 14:46 System.map-6.5.0-1020
252K -rw-r--r--  1 root root 251K Mar 25 17:52 config-6.5.0-1017-aws
252K -rw-r--r--  1 root root 251K May  1 14:46 config-6.5.0-1020-aws
 512 drwx------  3 root root  512 Jan  1  1970 efi
4.0K drwxr-xr-x  6 root root 4.0K May 15 06:13 grub
   0 lrwxrwxrwx  1 root root   25 May 15 06:13 initrd.img -> initrd.
 39M -rw-r--r--  1 root root  39M Apr 11 02:12 initrd.img-6.5.0-1017
 39M -rw-r--r--  1 root root  39M May 15 06:14 initrd.img-6.5.0-1020
```

/dev : Special or device files, which refer to physical devices such as hard disk, keyboard, monitor, mouse and modem etc

```
root@ip-172-31-36-123:/# cd dev
root@ip-172-31-36-123:/dev# ls -lsah
total 4.0K
   0 drwxr-xr-x 14 root root        3.3K May 30 04:47 .
4.0K drwxr-xr-x 19 root root        4.0K May 30 04:47 ..
   0 crw-r--r--  1 root root     10, 235 May 30 04:47 autofs
   0 drwxr-xr-x  2 root root         320 May 30 04:47 block
   0 crw-------  1 root root     10, 234 May 30 04:47 btrfs-control
   0 drwxr-xr-x  2 root root        2.6K May 30 04:47 char
   0 crw--w----  1 root tty       5,   1 May 30 04:47 console
   0 lrwxrwxrwx  1 root root          11 May 30 04:47 core -> /proc/kcore
   0 crw-------  1 root root     10, 123 May 30 04:47 cpu_dma_latency
   0 crw-------  1 root root     10, 203 May 30 04:47 cuse
   0 drwxr-xr-x  7 root root         140 May 30 04:47 disk
   0 drwxr-xr-x  2 root root          60 May 30 04:47 dma_heap
   0 crw-------  1 root root     10, 125 May 30 04:47 ecryptfs
   0 crw-rw----  1 root video    29,   0 May 30 04:47 fb0
   0 lrwxrwxrwx  1 root root          13 May 30 04:47 fd -> /proc/self/fd
   0 crw-rw-rw-  1 root root      1,   7 May 30 04:47 full
   0 crw-rw-rw-  1 root root     10, 229 May 30 04:47 fuse
```

/etc : Contains configuration files which are local to the machine. Some larger software packages, like Apache, can have their own subdirectories below /etc i.e. /etc/httpd. Some important subdirectories in /etc

```
root@ip-172-31-36-123:/# cd etc/
root@ip-172-31-36-123:/etc# ls -lsah
total 908K
4.0K drwxr-xr-x 100 root root      4.0K May 17 04:57 .
4.0K drwxr-xr-x  19 root root      4.0K May 30 04:47 ..
4.0K drwxr-xr-x   3 root root      4.0K May 15 03:30 .java
   0 -rw-------   1 root root         0 Apr 11 02:08 .pwd.lock
4.0K drwxr-xr-x   2 root root      4.0K Apr 11 02:09 PackageKit
4.0K drwxr-xr-x   8 root root      4.0K May 15 03:30 X11
4.0K drwxr-xr-x   4 root root      4.0K Apr 11 02:12 acpi
4.0K -rw-r--r--   1 root root      3.0K Apr 11 02:08 adduser.conf
4.0K drwxr-xr-x   2 root root      4.0K May 17 04:56 alternatives
4.0K drwxr-xr-x   8 root root      4.0K May 17 04:56 apache2
4.0K drwxr-xr-x   3 root root      4.0K Apr 11 02:09 apparmor
4.0K drwxr-xr-x   8 root root      4.0K May 17 04:55 apparmor.d
4.0K drwxr-xr-x   3 root root      4.0K Apr 11 02:09 apport
4.0K drwxr-xr-x   8 root root      4.0K May 15 03:38 apt
```

/home : Your sweet home to store data and other files. However in large installation the structure of /home directory depends on local administration decisions.

```
root@ip-172-31-36-123:/# cd home/
root@ip-172-31-36-123:/home# ls -lsah
total 12K
4.0K drwxr-xr-x  3 root   root   4.0K May 15 03:24 .
4.0K drwxr-xr-x 19 root   root   4.0K May 30 04:47 ..
4.0K drwxr-x---  5 ubuntu ubuntu 4.0K May 16 11:50 ubuntu
root@ip-172-31-36-123:/home#
```

/lib : This directory should hold those shared libraries that are necessary to boot the system and to run the commands in the root filesystem.

```
root@ip-172-31-36-123:/lib# ls
NetworkManager     debug              grub-legacy
X11                dpkg               hdparm
apache2            dracut             init
apparmor           ec2-hibinit-agent  initcpio
apt                environment.d      initramfs-tools
binfmt.d           file               jvm
byobu              finalrd            kernel
cgi-bin            firmware           klibc
cloud-init         girepository-1.0   klibc-BnzSoOUNgFnGkEcRdekugdBENMs.so
cnf-update-db      git-core           libdmmp.so
command-not-found  gnupg              libdmmp.so.0.2.0
```

/lib32 : 32 bit shared libraries that are necessary to boot the system and to run the commands in the root filesystem.

```
root@ip-172-31-36-123:/# cd lib32
root@ip-172-31-36-123:/lib32# ls -lsah
total 8.0K
4.0K drwxr-xr-x  2 root root 4.0K Apr 11 02:07 .
4.0K drwxr-xr-x 14 root root 4.0K Apr 11 02:07 ..
root@ip-172-31-36-123:/lib32#
```

/mnt : This directory contains mount points for temporarily mounted filesystems .

```
root@ip-172-31-36-123:/# cd mnt/
root@ip-172-31-36-123:/mnt# ls
root@ip-172-31-36-123:/mnt# ls -lsah
total 8.0K
4.0K drwxr-xr-x  2 root root 4.0K Apr 11 02:07 .
4.0K drwxr-xr-x 19 root root 4.0K May 30 04:47 ..
root@ip-172-31-36-123:/mnt#
```

/opt : This directory should contain add-on packages such as install download firefox or static files .

```
root@ip-172-31-36-123:/# cd opt/
root@ip-172-31-36-123:/opt# ls
root@ip-172-31-36-123:/opt# ls -lsah
total 8.0K
4.0K drwxr-xr-x  2 root root 4.0K Apr 11 02:07 .
4.0K drwxr-xr-x 19 root root 4.0K May 30 04:47 ..
root@ip-172-31-36-123:/opt#
```

/proc : This is a mount point for the proc filesystem, which provides information about running processes and the kernel.

```
root@ip-172-31-36-123:/# cd proc/
root@ip-172-31-36-123:/proc# ls -lsh
total 0
0 dr-xr-xr-x  9 root            root              0 May 30 04:47 1
0 dr-xr-xr-x  9 root            root              0 May 30 04:47 10
0 dr-xr-xr-x  9 ubuntu          ubuntu            0 May 30 05:17 1047
0 dr-xr-xr-x  9 ubuntu          ubuntu            0 May 30 05:17 1048
0 dr-xr-xr-x  9 root            root              0 May 30 04:48 1057
0 dr-xr-xr-x  9 root            root              0 May 30 05:17 1058
0 dr-xr-xr-x  9 root            root              0 May 30 04:48 1059
0 dr-xr-xr-x  9 root            root              0 May 30 05:17 1060
```

/root : This directory is usually the home directory for the root user.

```
root@ip-172-31-36-123:/# cd root/
root@ip-172-31-36-123:~# ls
snap
root@ip-172-31-36-123:~# cd snap/
root@ip-172-31-36-123:~/snap# ls -lsah
total 16K
4.0K drwx------ 4 root root 4.0K May 15 03:24 .
4.0K drwx------ 4 root root 4.0K May 29 05:47 ..
4.0K drwxr-xr-x 4 root root 4.0K May 15 03:24 amazon-ssm-agent
4.0K drwxr-xr-x 5 root root 4.0K May 16 03:55 lxd
root@ip-172-31-36-123:~/snap#
```

/sbin : Like /bin, this directory holds commands needed to boot the system, but which are usually not executed by normal users, root / admin user specific commands goes here.

```
root@ip-172-31-36-123:/# cd sbin/
root@ip-172-31-36-123:/sbin# ls -lsah
total 30M
 16K drwxr-xr-x  2 root root   16K May 17 04:56 .
4.0K drwxr-xr-x 14 root root  4.0K Apr 11 02:07 ..
   0 lrwxrwxrwx  1 root root    7 Apr 10 17:45 a2disconf -> a2enmod
   0 lrwxrwxrwx  1 root root    7 Apr 10 17:45 a2dismod -> a2enmod
   0 lrwxrwxrwx  1 root root    7 Apr 10 17:45 a2dissite -> a2enmod
   0 lrwxrwxrwx  1 root root    7 Apr 10 17:45 a2enconf -> a2enmod
```

/tmp : This directory contains temporary files which may be deleted with no notice, such as by a regular job or at system boot up.

```
root@ip-172-31-36-123:/# cd tmp/
root@ip-172-31-36-123:/tmp# ls
hsperfdata_jenkins                                          systemd-private-7a8f21449fdf4be78dd8356b33f508e4-chrony.service-NiuAJU
jetty-0_0_0_0-8080-war-_-any-2039300915510678694            systemd-private-7a8f21449fdf4be78dd8356b33f508e4-systemd-logind.service-wvtKDT
snap-private-tmp                                            systemd-private-7a8f21449fdf4be78dd8356b33f508e4-systemd-resolved.service-6kn9mA
systemd-private-7a8f21449fdf4be78dd8356b33f508e4-apache2.service-4x9KZE  winstone1263023304949537091 8.jar
root@ip-172-31-36-123:/tmp#
```

/usr : This directory is usually mounted from a separate partition. It should hold only sharable, read-only data, so that it can be mounted by various machines run ning Linux (useful for diskless client or multiuser Linux network such as university network). Programs, libraries, documentation etc. for all user-related programs.

```
root@ip-172-31-36-123:/usr# ls
bin  games  include  lib  lib32  lib64  libexec  libx32  local  sbin  share  src
root@ip-172-31-36-123:/usr#
```

/var : This directory contains files which may change in size, such as spool and log files.

```
bin  boot  dev  etc  home  lib  lib32  lib64  libx32  lost+found  media  mnt  opt  pr
root@ip-172-31-36-123:/# cd var/
root@ip-172-31-36-123:/var# ls
backups  cache  crash  lib  local  lock  log  mail  opt  run  snap  spool  tmp  www
root@ip-172-31-36-123:/var# cd www
root@ip-172-31-36-123:/var/www# ls
html
root@ip-172-31-36-123:/var/www# cd html/
root@ip-172-31-36-123:/var/www/html# ls
index.html
root@ip-172-31-36-123:/var/www/html# cat index.html
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/
<html xmlns="http://www.w3.org/1999/xhtml">
```

Common Linux log files name and usage

* /var/log/message: General message and system related stuff
* /var/log/auth.log: Authenication logs
* /var/log/kern.log: Kernel logs
* /var/log/cron.log: Crond logs (cron job)
* /var/log/maillog: Mail server logs
* /var/log/qmail/ : Qmail log directory (more files inside this directory)
* /var/log/httpd/: Apache access and error logs directory
* /var/log/lighttpd: Lighttpd access and error logs directory
* /var/log/boot.log : System boot log
* /var/log/mysqld.log: MySQL database server log file

* /var/log/secure: Authentication log
* /var/log/utmp or /var/log/wtmp : Login records file
* /var/log/yum.log: Yum log files

.How to get system information about a Linux server and change the hostname for that server.

CHECK SYSTEM INFO:

Kernel version:

```
[ec2-user@ip-172-31-24-25 ~]$ sudo su
[root@ip-172-31-24-25 ec2-user]# uname -r
5.10.216-204.855.amzn2.x86_64
```

uname -r

Operating System Details:

```
5.10.216-204.855.amzn2.x86_64
[root@ip-172-31-24-25 ec2-user]# cat /etc/os-release
NAME="Amazon Linux"
VERSION="2"
ID="amzn"
ID_LIKE="centos rhel fedora"
VERSION_ID="2"
PRETTY_NAME="Amazon Linux 2"
ANSI_COLOR="0;33"
CPE_NAME="cpe:2.3:o:amazon:amazon_linux:2"
HOME_URL="https://amazonlinux.com/"
SUPPORT_END="2025-06-30"
```

Cat /etc/os-release

```
[root@ip-172-31-24-25 ec2-user]# cat /proc/cpuinfo
processor       : 0
vendor_id       : GenuineIntel
cpu family      : 6
model           : 85
model name      : Intel(R) Xeon(R) Platinum 8259CL CPU @ 2.50GHz
stepping        : 7
microcode       : 0x5003605
cpu MHz         : 2499.992
cache size      : 36608 KB
physical id     : 0
siblings        : 2
core id         : 0
cpu cores       : 1
apicid          : 0
initial apicid  : 0
fpu             : yes
```

Cat /proc/cpuinfo - To get the CPU info.

```
[root@ip-172-31-24-25 ec2-user]# cat /proc/meminfo
MemTotal:         950076 kB
MemFree:          307200 kB
MemAvailable:     281268 kB
Buffers:               0 kB
Cached:            72700 kB
SwapCached:            0 kB
Active:             8956 kB
Inactive:         549812 kB
Active(anon):        348 kB
Inactive(anon):   486276 kB
Active(file):       8608 kB
```

cat /proc/meminfo - To get the memory information

Backup the existing hostname configuration:

```
DirectMap2M:      944128 kB
DirectMap1G:           0 kB
[root@ip-172-31-24-25 ec2-user]# sudo cp /etc/hostname /etc/hostname.backup
[root@ip-172-31-24-25 ec2-user]# sudo vi /etc/hostname
[root@ip-172-31-24-25 ec2-user]# sudo vi /etc/hostname
```

sudo cp /etc/hostname /etc/hostname.backup

Open the hostname file in a text editor:

```
[root@ip-172-31-24-25 ec2-user]# sudo cat /etc/hostname
bose-host
```

Update Hosts File:

Open the hosts file:

```
[root@ip-172-31-24-25 ec2-user]# sudo cat /etc/hostname
bose-host
[root@ip-172-31-24-25 ec2-user]# sudo cat /etc/hosts
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4 bose-host
::1         localhost6 localhost6.localdomain6
```

Apply the New Hostname:

Set the new hostname using the hostnamectl command:

sudo hostnamectl set-hostname bose-host

```
[root@ip-172-31-24-25 ec2-user]# sudo vi /etc/hosts
[root@ip-172-31-24-25 ec2-user]# sudo hostnamectl set-hostname bose-host
[root@ip-172-31-24-25 ec2-user]# sudo systemctl restart network
[root@ip-172-31-24-25 ec2-user]# sudo systemctl restart networking
```

Restart Networking Service:


```
[root@ip-172-31-24-25 ec2-user]# sudo vi /etc/hosts
[root@ip-172-31-24-25 ec2-user]# sudo hostnamectl set-hostname bose-host
[root@ip-172-31-24-25 ec2-user]# sudo systemctl restart network
[root@ip-172-31-24-25 ec2-user]# sudo systemctl restart networking
```


Verify the New Hostname:

```
Failed to restart networking.service: Unit not
[root@ip-172-31-24-25 ec2-user]# hostname
bose-host
```

Setting Up Partitions and Persistent Mounting on AWS EC2.

Objective:

Configure partitions on an AWS EC2 instance and ensure persistent mounting using /etc/fstab.

```
root@ip-172-31-36-123:/home/ubuntu# lsblk
NAME          MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0           7:0    0 25.2M  1 loop /snap/amazon-ssm-agent/7983
loop1           7:1    0 55.7M  1 loop /snap/core18/2812
loop2           7:2    0 55.7M  1 loop /snap/core18/2823
loop3           7:3    0 63.9M  1 loop /snap/core20/2264
loop4           7:4    0 63.9M  1 loop /snap/core20/2318
loop5           7:5    0   87M  1 loop /snap/lxd/27948
loop6           7:6    0   87M  1 loop /snap/lxd/28373
loop7           7:7    0 38.7M  1 loop /snap/snapd/21465
loop8           7:8    0 38.8M  1 loop /snap/snapd/21759
loop9           7:9    0 25.2M  1 loop /snap/amazon-ssm-agent/7993
nvme0n1       259:0    0    8G  0 disk
├─nvme0n1p1   259:1    0  7.9G  0 part /
├─nvme0n1p14  259:2    0    4M  0 part
└─nvme0n1p15  259:3    0  106M  0 part /boot/efi
nvme1n1       259:4    0    8G  0 disk
root@ip-172-31-36-123:/home/ubuntu# exit
exit
ubuntu@ip-172-31-36-123:~$ exit
```

Nvme1n1 is the newly attached volume . and we are going to make partitions on that.

Partition Creation:

Created multiple partitions on the instance using fdisk.
Assigned appropriate sizes to each partition (e.g., 2GB each).

```
Command (m for help): n
Partition type
   p   primary (1 primary, 0 extended, 3 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (2-4, default 2): 2
First sector (4196352-16777215, default 4196352):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (4196352-16777215, default 16777215): +2G

Created a new partition 2 of type 'Linux' and of size 2 GiB.

Command (m for help): n
Partition type
   p   primary (2 primary, 0 extended, 2 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (3,4, default 3): 3
First sector (8390656-16777215, default 8390656):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (8390656-16777215, default 16777215): +2G

Created a new partition 3 of type 'Linux' and of size 2 GiB.
```

```
   e   extended (container for logical partitions)
Select (default p): p
Partition number (3,4, default 3): 3
First sector (8390656-16777215, default 8390656):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (8390656-16777215, default 16777215): +2G

Created a new partition 3 of type 'Linux' and of size 2 GiB.

Command (m for help): n
Partition type
   p   primary (3 primary, 0 extended, 1 free)
   e   extended (container for logical partitions)
Select (default e):

Using default response e.
Selected partition 4
First sector (12584960-16777215, default 12584960):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (12584960-16777215, default 16777215):
```

Filesystem Formatting:

Formatted the partitions using mkfs.ext4 for ext4 filesystems.

> The mkfs command makes a new file system on a specified device. The mkfs command initializes the volume label, file system label, and startup block.

```
root@ip-172-31-36-123:/home/ubuntu# sudo mkfs.ext4 /dev/nvme1n1p1
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 524288 4k blocks and 131072 inodes
Filesystem UUID: fbd4e7f1-f26c-40a0-bb11-0d150a21805e
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

root@ip-172-31-36-123:/home/ubuntu# sudo mkfs.ext4 /dev/nvme1n1p2
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 524288 4k blocks and 131072 inodes
Filesystem UUID: 82055b32-a38b-4066-a41b-8e386e598b38
Superblock backups stored on blocks:
```

Updating /etc/fstab:

Edited the /etc/fstab file using sudo vi /etc/fstab.

Added entries for each partition using UUIDs to ensure persistent mounting.

Example entry: UUID=82055b32-a38b-4066-a41b-8e386e598b38 /mnt/data2 ext4 defaults,nofail 0 2.

```
root@ip-172-31-36-123:/home/ubuntu# sudo mkdir /mnt/data1
root@ip-172-31-36-123:/home/ubuntu# sudo mkdir /mnt/data2
root@ip-172-31-36-123:/home/ubuntu# sudo mkdir /mnt/data3
root@ip-172-31-36-123:/home/ubuntu# sudo mount /dev/nvme1n1p1 /mnt/data1
root@ip-172-31-36-123:/home/ubuntu# sudo mount /dev/nvme1n1p2 /mnt/data2
root@ip-172-31-36-123:/home/ubuntu#
^C
root@ip-172-31-36-123:/home/ubuntu# sudo mount /dev/nvme1n1p3 /mnt/data3
root@ip-172-31-36-123:/home/ubuntu# df -h
Filesystem       Size  Used Avail Use% Mounted on
```

Using sudo blkid command i got each partitions UUID's.

```
/dev/nvme0n1p1: LABEL="cloudimg-rootfs" UUID="db5e27f6-7377-40b0-9756-df259213cbb0" BLOCK_SIZE="4096" TYPE="ext4"
/dev/nvme0n1p15: LABEL_FATBOOT="UEFI" LABEL="UEFI" UUID="692D-8804" BLOCK_SIZE="512" TYPE="vfat" PARTUUID="e556a7
/dev/loop1: TYPE="squashfs"
/dev/nvme0n1p14: PARTUUID="d5a642d2-c9d7-4be1-a293-bcc064177b4f"
/dev/loop8: TYPE="squashfs"
/dev/loop6: TYPE="squashfs"
/dev/loop4: TYPE="squashfs"
/dev/loop2: TYPE="squashfs"
/dev/loop0: TYPE="squashfs"
/dev/loop9: TYPE="squashfs"
/dev/loop7: TYPE="squashfs"
/dev/nvme1n1p2: UUID="82055b32-a38b-4066-a41b-8e386e598b38" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="3530a934-02"
/dev/nvme1n1p3: UUID="4cce72ce-6750-46a2-89b6-8f6ab12fdc17" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="3530a934-03"
/dev/nvme1n1p1: UUID="fbd4e7f1-f26c-40a0-bb11-0d150a21805e" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="3530a934-01"
/dev/loop5: TYPE="squashfs"
```

As you can see i have added UUID in /etc/fstab

```
tmpfs           383M  4.0K  383M   1% /run/user/1000
root@ip-172-31-36-123:/home/ubuntu# sudo cat /etc/fstab
LABEL=cloudimg-rootfs   /        ext4   discard,errors=remount-ro       0 1
LABEL=UEFI      /boot/efi       vfat    umask=0077      0 1

UUID=82055b32-a38b-4066-a41b-8e386e598b38 /mnt/data2 ext4 defaults,nofail 0 2
UUID=4cce72ce-6750-46a2-89b6-8f6ab12fdc17 /mnt/data3 ext4 defaults,nofail 0 2
UUID=fbd4e7f1-f26c-40a0-bb11-0d150a21805e /mnt/data1 ext4 defaults,nofail 0 2
root@ip-172-31-36-123:/home/ubuntu#
```

After all i did sudo mount -a for checking any errors and rebooted my instance.

Mounting Verification:

Used sudo mount -a to test the /etc/fstab entries for errors.
Verified successful mounting using df -h to check mounted filesystems.

```
Last login: Fri May 31 06:09:26 2024 from 27.6.32.62
ubuntu@ip-172-31-36-123:~$ sudo su
root@ip-172-31-36-123:/home/ubuntu# df -h
Filesystem       Size  Used Avail Use% Mounted on
/dev/root        7.6G  3.6G  4.0G  48% /
tmpfs            1.9G     0  1.9G   0% /dev/shm
tmpfs            766M  896K  765M   1% /run
tmpfs            5.0M     0  5.0M   0% /run/lock
efivarfs         128K  4.1K  119K   4% /sys/firmware/efi/efivars
/dev/nvme1n1p3   2.0G   24K  1.8G   1% /mnt/data3
/dev/nvme1n1p2   2.0G   24K  1.8G   1% /mnt/data2
/dev/nvme0n1p15  105M  6.1M   99M   6% /boot/efi
/dev/nvme1n1p1   2.0G   24K  1.8G   1% /mnt/data1
tmpfs            383M  4.0K  383M   1% /run/user/1000
root@ip-172-31-36-123:/home/ubuntu# sudo cat /etc/fstab
LABEL=cloudimg-rootfs   /        ext4   discard,errors=remount-ro
LABEL=UEFI      /boot/efi       vfat    umask=0077      0 1

UUID=82055b32-a38b-4066-a41b-8e386e598b38 /mnt/data2 ext4 defaults,
```

System Reboot Testing:

Rebooted the EC2 instance to verify that the mounts persist after a reboot.
Checked mounts again post-reboot to ensure persistence.

```
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Fri May 31 06:09:26 2024 from 27.6.32.62
ubuntu@ip-172-31-36-123:~$ sudo su
root@ip-172-31-36-123:/home/ubuntu# df -h
Filesystem       Size  Used Avail Use% Mounted on
/dev/root        7.6G  3.6G  4.0G  48% /
tmpfs            1.9G     0  1.9G   0% /dev/shm
tmpfs            766M  896K  765M   1% /run
tmpfs            5.0M     0  5.0M   0% /run/lock
efivarfs         128K  4.1K  119K   4% /sys/firmware/efi/efivars
/dev/nvme1n1p3   2.0G   24K  1.8G   1% /mnt/data3
/dev/nvme1n1p2   2.0G   24K  1.8G   1% /mnt/data2
/dev/nvme0n1p15  105M  6.1M   99M   6% /boot/efi
/dev/nvme1n1p1   2.0G   24K  1.8G   1% /mnt/data1
tmpfs            383M  4.0K  383M   1% /run/user/1000
root@ip-172-31-36-123:/home/ubuntu# sudo cat /etc/fstab
LABEL=cloudimg-rootfs   /        ext4   discard,errors=remount-ro
LABEL=UEFI      /boot/efi       vfat    umask=0077      0 1
```

Final Verification:

Confirmed that all partitions are correctly mounted and usable post-reboot.
Ensured no errors or issues in the mount configurations.

Create an SSH user and connect using both key-based and password-based authentication. Explore SSH configurations.

SSH Key Generation:

To generate an SSH key pair, we used the command `ssh-keygen -t rsa -b 4096 -C "your_email@example.com"`. This command generates an RSA key pair with a bit length of 4096 and associates it with the specified email address.

```
$ ls -l /c/Users/Subash.Murugan/.ssh/id_rsa
-rw-r--r-- 1 subash.murugan 1049089 3401 May 30 11:35 /c/Users/Subash.Murugan/.s
sh/id_rsa

subash.murugan@PWSLTP178 MINGW64 ~
$ cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAACAQCvPMJMznGl5/kR3GX4czzMyY/ILQ9fSniJ19babHx3
Okis4I4XSukQK2yLfp0DAjIJ+qJdKsa/AxNDAH2KuKf7eO81ehciNBKf3JNAOfN+XkrMZ2dBsWZ/zInw
v367JP198bm9hP6p+9b0jt5D7Skj3he93Rwsmw8YXhJIIiU/Fi0K1M/mkKz764dz+m2cDEkNgQ09pfo6
6K+C/0/aAldL82De5IT2yXtKmmmo/5mJaYKVAKpi2YKiVC8VDi/ypn7BN4GUoGp7aiCLGLfWVHEg8My0
CGUZ/mE4OUqwTq8iGNLDfbJkDV3ycAtls2zfRnPnM8hF5+dRa3nqSmPe32rt/y7KPsjBddFaE8I1naEg
HsOUVi2A4yQWL4YmLYMbHSwbTscEZKM5b577E4rsppm3PiPke5E1h2PNIPr8Lh5dc+jahgWbcHmZ471A
E7s89/gxEtga+oOY9rCKhfV5pkpFTfqAf8SEbv+7R/nDuyemVCJT1yNnEP5OX1H5RrZWcDZOtrS6QI7V
hitd2O8ga2Orhzx5vXOp/vPh247P9H7XsoSmNZTXu0y0Dbp/CKg/aAPy5vzEYCA9Ieu1YrUtKdDLiVgc
4IZUxVMU2zMDhAwtcnLsW7xJ/vASF82YS5mYmg8n8hd/7rExGqiEnF2uTx/xCGUrYU1LpdnWnYUQwJ2/
Uw== subashchandra@karunya.edu.in

subash.murugan@PWSLTP178 MINGW64 ~
$ cd .ssh/

subash.murugan@PWSLTP178 MINGW64 ~/.ssh
$ ls
id_rsa  id_rsa.pub  known_hosts  known_hosts.old
```

Copying Public Key to Remote Server:

```
subhash@ip-172-31-36-123:~$ sudo su
[sudo] password for subhash:
root@ip-172-31-36-123:/home/subhash# cat ~/.ssh/authorized_keys
no-port-forwarding,no-agent-forwarding,no-X11-forwarding,command="echo 'Please login as the user \"ubuntu\" rather than the user \"root\".';echo;sleep 10;exit 142" s
-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQCntNzbrqc+h57fmjGa+u3/9u63BMB5qMEiF9mI1So62BYuXaYB7g/fpgBkWPq1xgi/qqwOBasEpJY1trV7SCdahUKhiADEffQlOonNM5RWmetMSHdWiuSQmNiMrk6G1Vs
HcFn4WdMyPJfBqiEadg7LvSxmHgH4LnBO99uuAJycsYdQCFY5O1WrV7XR5Yp/fVCd3/SyZDW/uHOcd3Wv2atVOjpV6vLFgn/uu9aJIONSjOgWh5/USWin1HJZ3v/33i8hmtlDegHTbVy3xtq4IUtA/4B7IiXBOiPLYAc
WF85+O4fX4feIk4wihNh6j6TwQ+iuZa1W+ifh9DYXi3 subhash
root@ip-172-31-36-123:/home/subhash#
```

Check File Permissions:

Ensure that the permissions for the `~/.ssh` **directory and the** `authorized_keys` **file are correct:**

chmod 700 ~/.ssh
chmod 600 ~/.ssh/authorized_keys

Ensure Ownership:

Verify that the `~/.ssh` **directory and** `authorized_keys` **file are owned by the** `subhash` **user:**

chown -R subhash:subhash ~/.ssh

Verify SSH Configuration

Make sure your `sshd_config` on the server allows for key-based authentication. Check the following settings in `/etc/ssh/sshd_config`:

```
# Authentication:

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

PubkeyAuthentication no

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile     .ssh/authorized_keys .ssh/authorized_keys2
```

```
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication yes
#PermitEmptyPasswords no

# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
KbdInteractiveAuthentication no
```

I did pubkeyauthentication no and configured password authentication as yes so that it prompt to ask password on each login.

```
subash.murugan@PWSLTP178 MINGW64 ~/Downloads
$ ssh subhash@ec2-13-53-134-40.eu-north-1.compute.amazonaws.com
subhash@ec2-13-53-134-40.eu-north-1.compute.amazonaws.com's password:
Last login: Mon Jun  3 06:46:42 2024 from 196.12.58.69
subhash@ip-172-31-36-123:~$
```

Now i changed pubkeyauthentication as yes and checked again it doesnt ask for password since local machine keypairs are configured.

```
root@ip-172-31-36-123:/home/subhash# sudo vi /etc/ssh/sshd_config
root@ip-172-31-36-123:/home/subhash# sudo systemctl restart sshd
root@ip-172-31-36-123:/home/subhash#
exit
subhash@ip-172-31-36-123:~$
logout
Connection to ec2-13-53-134-40.eu-north-1.compute.amazonaws.com closed.

subash.murugan@PWSLTP178 MINGW64 ~/Downloads
$ ssh subhash@ec2-13-53-134-40.eu-north-1.compute.amazonaws.com
Last login: Mon Jun  3 07:37:32 2024 from 196.12.58.69
subhash@ip-172-31-36-123:~$ sudo su
```

This time it does not asked for password.

And also i have tried login with different machine and then it asked for password.Since other machine local key pairs are not configured.

```
yaswanth@PWSLTP204:~$ sudo ssh subash@13.53.134.40
The authenticity of host '13.53.134.40 (13.53.134.40)' can't be established.
ED25519 key fingerprint is SHA256:OyVOAMXn2RqQMfAu23HMnQcLDgN95qKI19tZAk78pOE.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.53.134.40' (ED25519) to the list of known hosts.
subash@13.53.134.40's password:

yaswanth@PWSLTP204:~$
```

---

work on scp and rsync commands for copy files from one server to another server.

Local machine to EC2 Instance:

```
subash.murugan@PWSLTP178 MINGW64 ~/Downloads
$ scp -i subhash.pem MySQLNotesForProfessionals.pdf subhash.murugan@ec2-13-53-134-40.eu-north-1.compute.amazonaws.com:/home/subhash/sqlnotes/
subhash.murugan@ec2-13-53-134-40.eu-north-1.compute.amazonaws.com's password:

subash.murugan@PWSLTP178 MINGW64 ~/Downloads
$ scp -i subhash.pem MySQLNotesForProfessionals.pdf subhash@ec2-13-53-134-40.eu-north-1.compute.amazonaws.com:/home/subhash/sqlnotes/
subhash@ec2-13-53-134-40.eu-north-1.compute.amazonaws.com's password:
MySQLNotesForProfessionals.pdf                                    100% 1920KB 980.5KB/s   00:01
```

```
/home/subhash/sqlnotes
subhash@ip-172-31-36-123:~/sqlnotes$ ls
MySQLNotesForProfessionals.pdf
subhash@ip-172-31-36-123:~/sqlnotes$
```

As you can see the file is copied successfully from local machine to cloud server.

$ scp -i subhash.pem MySQLNotesForProfessionals.pdf subhash@ec2-13-53-134-40.eu-north-1.compute.amazonaws.com:/home/subhash/sqlnotes/

Work on installation of lamp with latest version and change mysql root directory.

Change MySQL Data Directory:

Stop MySQL service:

systemctl stop mysql

```
root@ip-172-31-36-123:/home/subhash# systemctl stop mysql
root@ip-172-31-36-123:/home/subhash# mkdir -p /new/mysql/datadir
```

Create the new data directory (e.g., `/new/mysql/datadir`):

```
Unit mariadb.service could not be found.
root@ip-172-31-36-123:/home/subhash# systemctl stop mysql
root@ip-172-31-36-123:/home/subhash# mkdir -p /new/mysql/datadir
```

mkdir -p /new/mysql/datadir

Copy the existing data to the new directory:

```
root@ip-172-31-36-123:/home/subhash# cp -R /var/lib/mysql/* /new/mysql/datadir/
root@ip-172-31-36-123:/home/subhash# chown -R mysql:mysql /new/mysql/datadir
```

cp -R /var/lib/mysql/* /new/mysql/datadir/

Change ownership of the new data directory:

chown -R mysql:mysql /new/mysql/datadir

Edit the MySQL configuration file to point to the new data directory:

vi /etc/mysql/mysql.conf.d/mysqld.cnf

Find the `datadir` line and update it to:

datadir = /new/mysql/datadir

```
# Basic Settings
#
user            = mysql
# pid-file      = /var/run/mysqld/mysqld.pid
# socket        = /var/run/mysqld/mysqld.sock
# port          = 3306
datadir         = /new/mysql/datadir
```

Update the AppArmor configuration for MySQL (if AppArmor is enabled):

vi /etc/apparmor.d/tunables/alias

Add or update the line:

alias /var/lib/mysql/ -> /new/mysql/datadir/,

```
# Alias rules can be used to rewrite paths and are done after
# resolution. For example, if '/usr' is on removable media:
# alias /usr/ -> /mnt/usr/,
#
# Or if mysql databases are stored in /home:
# alias /var/lib/mysql/ -> /home/mysql/,
alias /var/lib/mysql/ -> /new/mysql/datadir/,
```

Reload AppArmor profiles:

systemctl reload apparmor

Start MySQL service:

systemctl start mysql

Log into MySQL:

mysql -u root -p

Check the current data directory:

SHOW VARIABLES LIKE 'datadir';

```
affiliates. Other names may be trademarks of their res
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the

mysql> SHOW VARIABLES LIKE 'datadir';
+---------------+---------------------+
| Variable_name | Value               |
+---------------+---------------------+
| datadir       | /new/mysql/datadir/ |
+---------------+---------------------+
1 row in set (0.00 sec)

mysql>
```

Work on linux users creation & deletion with different use cases, like modify the possible ways for user related properties.

User Creation:

```
root@ip-172-31-36-123:/home/ubuntu#  sudo su
root@ip-172-31-36-123:/home/ubuntu# sudo adduser bose
Adding user `bose' ...
Adding new group `bose' (1003) ...
Adding new user `bose' (1003) with group `bose' ...
Creating home directory `/home/bose' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for bose
Enter the new value, or press ENTER for the default
```

Change user's home directory:

```
root@ip-172-31-36-123:/home/ubuntu# sudo usermod -d /new/home/directory bose
root@ip-172-31-36-123:/home/ubuntu# cd ..
root@ip-172-31-36-123:/home# ls
bose  magento  subhash  ubuntu
root@ip-172-31-36-123:/home#
```

Create group and add users to a group:

```
root@ip-172-31-36-123:/home/bose# sudo groupadd mygroup
root@ip-172-31-36-123:/home/bose# sudo adduser bose
adduser: The user `bose' already exists.
root@ip-172-31-36-123:/home/bose# sudo usermod -aG mygroup bose
root@ip-172-31-36-123:/home/bose#  groups bose
bose : bose mygroup
```

Grant read and write permissions to the owner:

```
bosworks
root@ip-172-31-36-123:/home/bose# chmod u+rw bosworks
```

Remove execute permission for the group:

```
root@ip-172-31-36-123:/home/bose# ls
bosworks
root@ip-172-31-36-123:/home/bose# chmod g-x bosworks
```

Set permissions to read-only for others:

```
chmod: cannot access 'filename': No such file or direct
root@ip-172-31-36-123:/home/bose# chmod o=r bosworks/
root@ip-172-31-36-123:/home/bose# ^C
root@ip-172-31-36-123:/home/bose# ls
```

Set permissions to `rwxr-xr--`:

Understanding chmod 754
The numeric mode 754 represents specific permissions for the owner, group, and others:

7 (owner): Read (4) + Write (2) + Execute (1) = 7
5 (group): Read (4) + Execute (1) = 5
4 (others): Read (4) = 4
This means:

The owner can read, write, and execute the file/directory.
The group can read and execute the file/directory.
Others can only read the file/directory.

```
root@ip-172-31-36-123:/home/bose# ls -ld bosworks
drwxr-xr-- 2 root root 4096 Jun  7 06:58 bosworks
root@ip-172-31-36-123:/home/bose#
```

In UNIX chmod 1 = execute, read is 4 and write is 2.

```
Try 'chmod --help' for more information.
root@ip-172-31-36-123:/home/bose# chmod --help
Usage: chmod [OPTION]... MODE[,MODE]... FILE...
  or:  chmod [OPTION]... OCTAL-MODE FILE...
  or:  chmod [OPTION]... --reference=RFILE FILE...
Change the mode of each FILE to MODE.
With --reference, change the mode of each FILE to that of RFILE.

  -c, --changes          like verbose but report only when a change is made
  -f, --silent, --quiet  suppress most error messages
  -v, --verbose          output a diagnostic for every file processed
      --no-preserve-root  do not treat '/' specially (the default)
      --preserve-root    fail to operate recursively on '/'
      --reference=RFILE  use RFILE's mode instead of MODE values
  -R, --recursive        change files and directories recursively
      --help     display this help and exit
      --version  output version information and exit

Each MODE is of the form '[ugoa]*([-+=]([rwxXst]*|[ugo]))+|[-+=][0-7]+'.

GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
Report any translation bugs to <https://translationproject.org/team/>
```

The chmod 664 command sets the permissions for a file or directory. Here's a detailed breakdown of what chmod 664 means and how to use it:

Understanding chmod 664
The numeric mode 664 represents specific permissions for the owner, group, and others:

6 (owner): Read (4) + Write (2) = 6
6 (group): Read (4) + Write (2) = 6
4 (others): Read (4) = 4
This means:

The owner can read and write the file/directory.
The group can read and write the file/directory.
Others can only read the file/directory.

.How to get system information about a Linux server and change the hostname for that server.

CHECK SYSTEM INFO:

Kernel version:

```
[ec2-user@ip-172-31-24-25 ~]$ sudo su
[root@ip-172-31-24-25 ec2-user]# uname -r
5.10.216-204.855.amzn2.x86_64
```

uname -r

Operating System Details:

```
5.10.216-204.855.amzn2.x86_64
[root@ip-172-31-24-25 ec2-user]# cat /etc/os-release
NAME="Amazon Linux"
VERSION="2"
ID="amzn"
ID_LIKE="centos rhel fedora"
VERSION_ID="2"
PRETTY_NAME="Amazon Linux 2"
ANSI_COLOR="0;33"
CPE_NAME="cpe:2.3:o:amazon:amazon_linux:2"
HOME_URL="https://amazonlinux.com/"
SUPPORT_END="2025-06-30"
```

Cat /etc/os-release

```
[root@ip-172-31-24-25 ec2-user]# cat /proc/cpuinfo
processor       : 0
vendor_id       : GenuineIntel
cpu family      : 6
model           : 85
model name      : Intel(R) Xeon(R) Platinum 8259CL CPU @ 2.50GHz
stepping        : 7
microcode       : 0x5003605
cpu MHz         : 2499.992
cache size      : 36608 KB
physical id     : 0
siblings        : 2
core id         : 0
cpu cores       : 1
apicid          : 0
initial apicid  : 0
fpu             : yes
```

Cat /proc/cpuinfo - To get the CPU info.

```
[root@ip-172-31-24-25 ec2-user]# cat /proc/meminfo
MemTotal:         950076 kB
MemFree:          307200 kB
MemAvailable:     281268 kB
Buffers:               0 kB
Cached:            72700 kB
SwapCached:            0 kB
Active:             8956 kB
Inactive:         549812 kB
Active(anon):        348 kB
Inactive(anon):   486276 kB
Active(file):       8608 kB
```

cat /proc/meminfo - To get the memory information

Backup the existing hostname configuration:

```
DirectMap2M:      944128 kB
DirectMap1G:           0 kB
[root@ip-172-31-24-25 ec2-user]# sudo cp /etc/hostname /etc/hostname.backup
[root@ip-172-31-24-25 ec2-user]# sudo vi /etc/hostname
[root@ip-172-31-24-25 ec2-user]# sudo vi /etc/hostname
```

sudo cp /etc/hostname /etc/hostname.backup

Open the hostname file in a text editor:

```
[root@ip-172-31-24-25 ec2-user]# sudo cat /etc/hostname
bose-host
```

Update Hosts File:

Open the hosts file:

```
[root@ip-172-31-24-25 ec2-user]# sudo cat /etc/hostname
bose-host
[root@ip-172-31-24-25 ec2-user]# sudo cat /etc/hosts
127.0.0.1   localhost localhost.localdomain localhost4 localhost4.localdomain4 bose-host
::1         localhost6 localhost6.localdomain6
```

Apply the New Hostname:

Set the new hostname using the hostnamectl command:

sudo hostnamectl set-hostname bose-host

```
[root@ip-172-31-24-25 ec2-user]# sudo vi /etc/hosts
[root@ip-172-31-24-25 ec2-user]# sudo hostnamectl set-hostname bose-host
[root@ip-172-31-24-25 ec2-user]# sudo systemctl restart network
[root@ip-172-31-24-25 ec2-user]# sudo systemctl restart networking
```

Restart Networking Service:


```
[root@ip-172-31-24-25 ec2-user]# sudo vi /etc/hosts
[root@ip-172-31-24-25 ec2-user]# sudo hostnamectl set-hostname bose-host
[root@ip-172-31-24-25 ec2-user]# sudo systemctl restart network
[root@ip-172-31-24-25 ec2-user]# sudo systemctl restart networking
```


Verify the New Hostname:

```
Failed to restart networking.service: Unit not
[root@ip-172-31-24-25 ec2-user]# hostname
bose-host
```